

Code-Layout, Readability and Reusability

Date	15 March 2024
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Used standard Python indentation (4 spaces).
2	Proper File Structure	Files and folders are logically organized	Yes	Project organized into dataset, model, templates, static, and application files.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables such as income, credit_history, and prediction_result are descriptive
4	Function / Method Names	Functions are descriptively named	Yes	Functions like train_model(), predict_approval(), and load_model() are self-explanatory.
5	Code Comments	Inline and block comments explain logic	Partial	Comments added for important sections and model logic.
6	Modular Design	Code is split into reusable functions/modules	Yes	Separate modules used for preprocessing, training, prediction, and Flask application.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Repeated code minimized using reusable functions.
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic input validation and exception handling implemented.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technolog	Where Reused	Reusability Level (High / Medium / Low)
1	Data Preprocessing Module	Python (Pandas)	Data cleaning and feature preparation	High

2	Machine Learning Model	Scikit-learn / XGBoost	Credit card approval prediction	High
3	Flask Application	Python (Flask)	HTML, CSS, Bootstrap	High
4	Database Module	MySQL	Applicant information storage	Medium
5	User Interface	HTML, CSS, Bootstrap	Input forms and result display	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure with separate modules.
Readability	5	Meaningful names and consistent formatting improve readability.
Reusability	5	Modular components can be reused in future ML projects.
Documentation / Comments	4	Comments provided for important sections; additional documentation can be added.
Overall Score	4.8	The project follows good coding practices and is maintainable.